

PROFILE

Software developer and independent builder with practical experience developing web applications, backend systems, AI-powered software, IoT systems, and developer tools. Experienced in taking ideas from concept through implementation and debugging, with a strong ability to learn unfamiliar technologies and work across both software and hardware systems.

01 SELECTED PROJECTS

Dave — AI Personal Assistant

AI / SYSTEMS

Python · Gemini API · ChromaDB · Embeddings · Docker

Designed an AI assistant with persistent semantic memory, using vector embeddings and ChromaDB for context retrieval. Explored modular architecture, speech interfaces, and containerized development with Docker and WSL.

AgriFlow — Automated Irrigation System

IOT / AUTOMATION

Raspberry Pi · Python · GPIO · MQTT · Sensors

Built a hardware/software irrigation system that combines environmental sensing, Raspberry Pi control, automation logic, and MQTT communication to connect software decisions with physical system behavior.

WeaverBird IDE

DEVELOPER TOOLS

JavaScript · HTML · CSS · Web APIs

Built a browser-based development environment with custom coding workflows and desktop-style application architecture, exploring how development tooling can be implemented directly within the web platform.

School Report Card System

WEB / PDF

JavaScript · PDF.js · Document Processing

Developed a browser-based document workflow for editing school report cards, including coordinate-based placement of information over PDF templates.

School Prep System (SPS)

EDUCATION

JavaScript · Backend Development · Databases

Developed an educational application supporting student preparation workflows with frontend, backend, and database-backed functionality.

WebOS / WebOSUIKit

UI ENGINEERING

JavaScript · HTML · CSS

Built reusable components for a desktop-like operating environment inside the browser, exploring window management, application interfaces, and reusable frontend architecture.

Raspberry Pi & Computer Vision Projects

EMBEDDED

Python · OpenCV · Picamera2 · GPIO

Developed hardware/software experiments involving camera systems, computer vision, sensors, GPIO, and Python while troubleshooting integration across physical and software layers.

02 EXPERIENCE

Independent Software Developer

Project-Based

Software · AI · IoT · Product Development

- Design and build software projects from initial concept through architecture, implementation, debugging, and iteration.
- Work across frontend, backend, databases, APIs, AI integrations, hardware interfaces, and deployment.
- Independently research and adopt unfamiliar technologies when required by project goals.

TECHNICAL SKILLS

Languages

Python, JavaScript, PHP, Kotlin, SQL, HTML, CSS

Frontend

Vue.js, Tailwind CSS

Backend

FastAPI, Node.js, Express.js, Fastify

Databases

MySQL, PostgreSQL, MongoDB

AI & Data

Gemini API, embeddings, ChromaDB, semantic search

Tools

Git, GitHub, Docker, WSL, Vercel, VS Code

Hardware

Raspberry Pi, Arduino, GPIO, MQTT, OpenCV, sensors

EDUCATION

Tema Senior High School

General Science with Elective ICT

GRADUATED · 2026

AREAS

AI Systems

Full-Stack

Robotics

IoT

Automation

Developer Tools

UI/UX

Product

LANGUAGE

English · Fluent